Project Name

# Food Waste Management System


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# Workbook Template

## Project Name:

Food Waste Management System

A Digital Platform to Connect Surplus Food Donors with Those in Need

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## Problem Statement:

Large quantities of edible food are discarded every day by restaurants, hotels, caterers and events, while many people go without enough to eat. Surplus food is wasted mainly because no efficient channel links those who have extra food with those who need it. Manual coordination is slow, untracked and leads to spoilage, missed pickups and no data on how much food is donated, collected or wasted. The Food Waste Management System provides a digital platform where donors list surplus food, NGOs / receivers request it, volunteers collect and deliver it, and an admin monitors activity, waste and reports. Food that is no longer fit for use is recorded, categorized and sent for composting or recycling.

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## Why does this problem exist?

1. Lack of a proper channel between donors and receivers
2. Poor communication of surplus food information
3. Manual coordination through phone calls and personal contacts
4. No tracking of donations after they are made
5. Spoilage and missed pickups of time-sensitive food
6. Lack of coordination between donors, NGOs and volunteers
7. No records of food donated, collected or wasted

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## Who are the stakeholders?

Internal users:

1. Food Waste Administrator
2. Volunteers
3. Organization / Institute Management
4. System Administrator / Developer

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External users:

1. Donors (restaurants, hotels, caterers, event organizers)
2. NGOs / Receivers
3. Composting / Recycling Agencies
4. Government / Local Authorities
5. Community Members (beneficiaries)

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## Expected system users:

1. Administrator
2. Donors
3. NGOs / Receivers

4. Volunteers

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## Objectives of the project:

1. To develop an online food waste management system.
2. To let donors quickly list surplus food with quantity, location and expiry time.
3. To let NGOs / receivers view available food and request it.
4. To assign volunteers to collect and deliver food to receivers.
5. To record and categorize waste, and manage composting / recycling.
6. To give the admin tools to manage users and donations and monitor monthly / yearly reports.
7. To maintain accurate donation, delivery and waste records.
8. To reduce food spoilage and wastage.

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## Expected outputs:

1. Available surplus food listings.
2. Online food request and pickup assignment.
3. Donation details with quantity, location and expiry time.
4. Donation status such as Available, Assigned, Delivered.
5. Notifications and alerts on donation status.
6. Donor donation history.
7. Admin dashboard.
8. Monthly and yearly reports of donated, collected and wasted food.

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## Hands-on Activity

Identify:

✓ **Strategic information:**

1. Analyze long-term food donation and waste trends.
2. Plan the number of volunteers and NGO partnerships needed.
3. Improve services based on monthly / yearly reports.

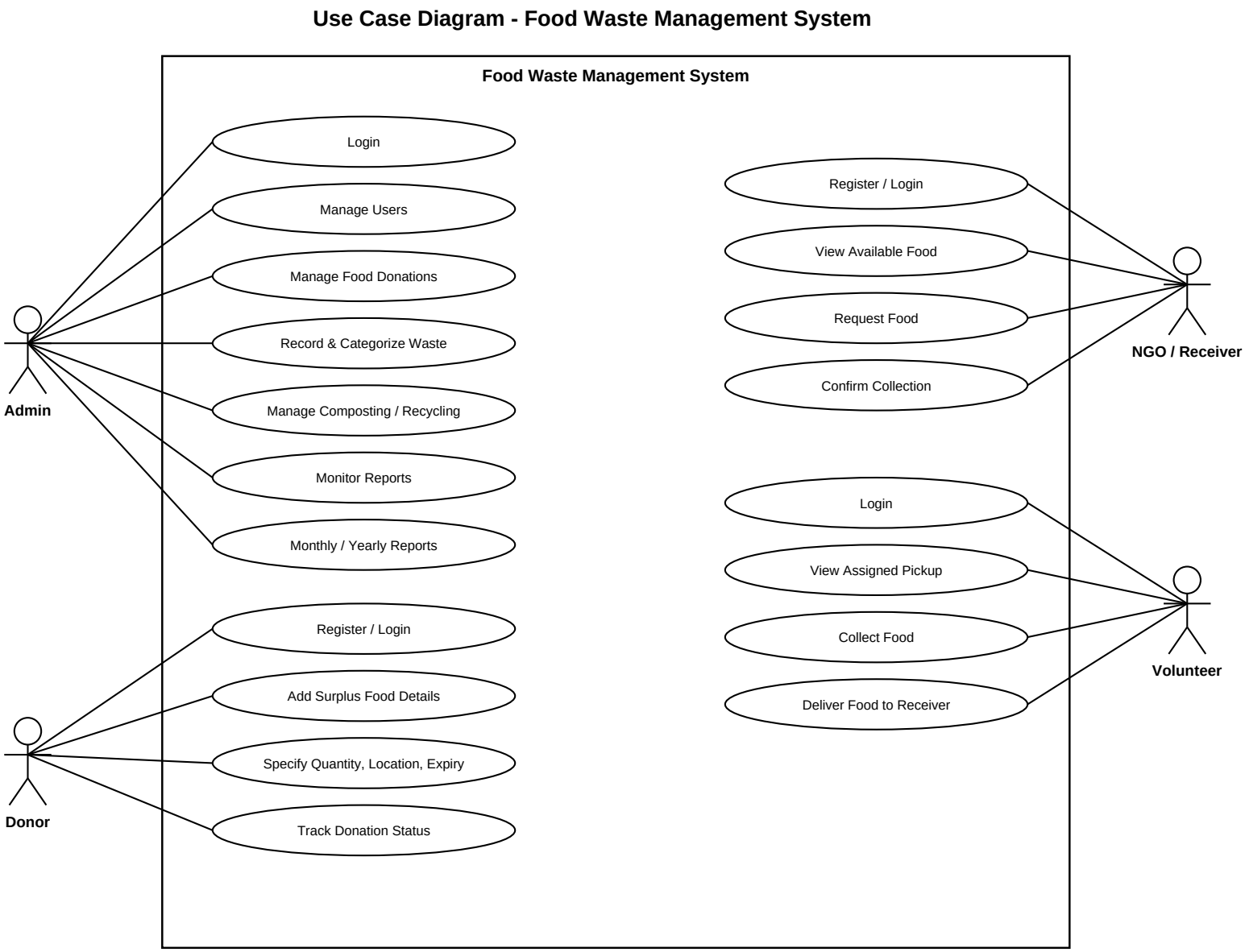
✓ **Tactical information:**

1. Schedule volunteer pickups.
2. Map donors and NGOs by area.
3. Manage composting / recycling workload.

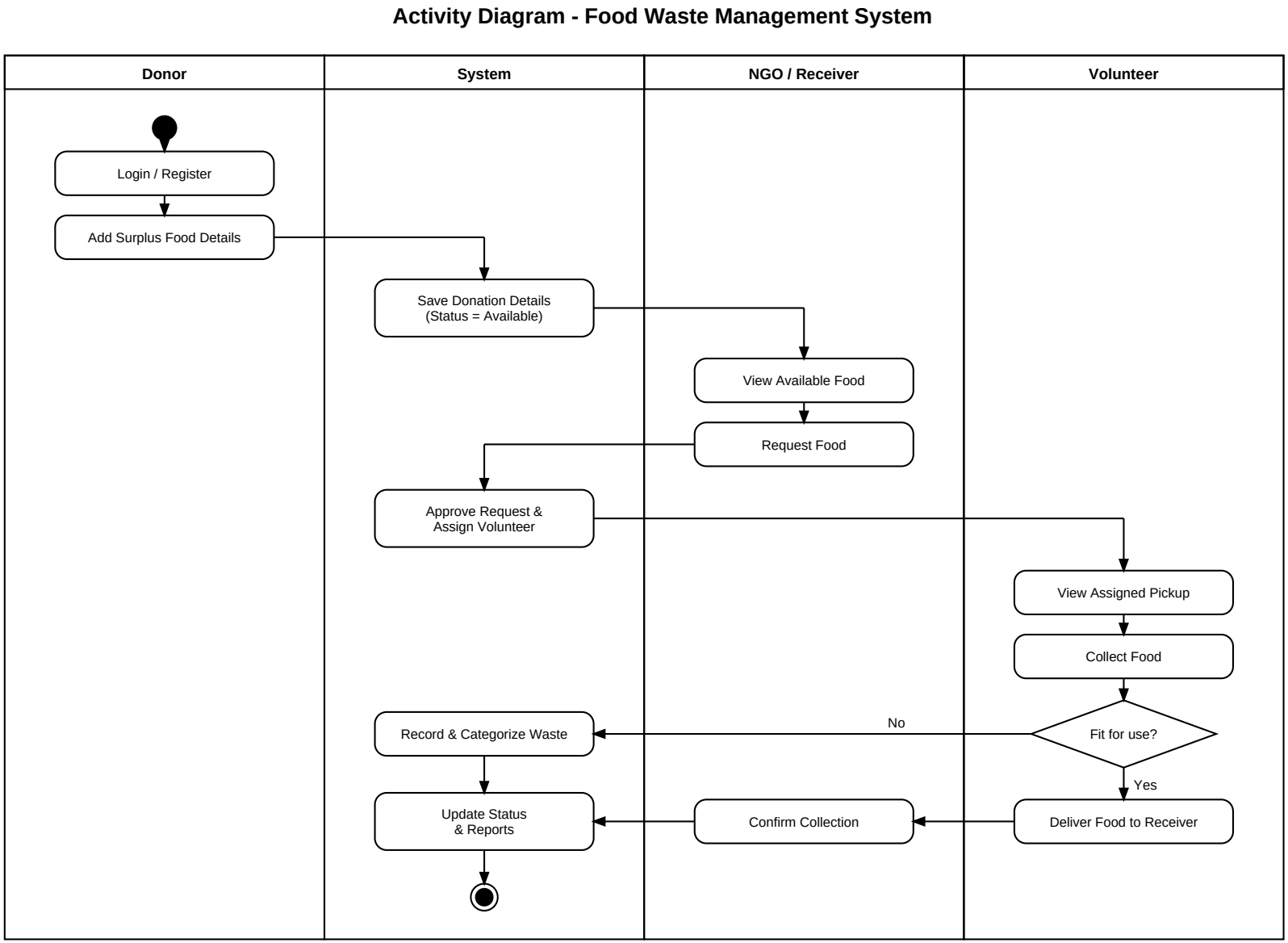
✓ **Operational information:**

1. Add surplus food details (quantity, location, expiry).
2. Approve food requests and assign a volunteer.
3. Confirm pickup, delivery and collection, and update donation status.
4. Record and categorize unfit food as waste.

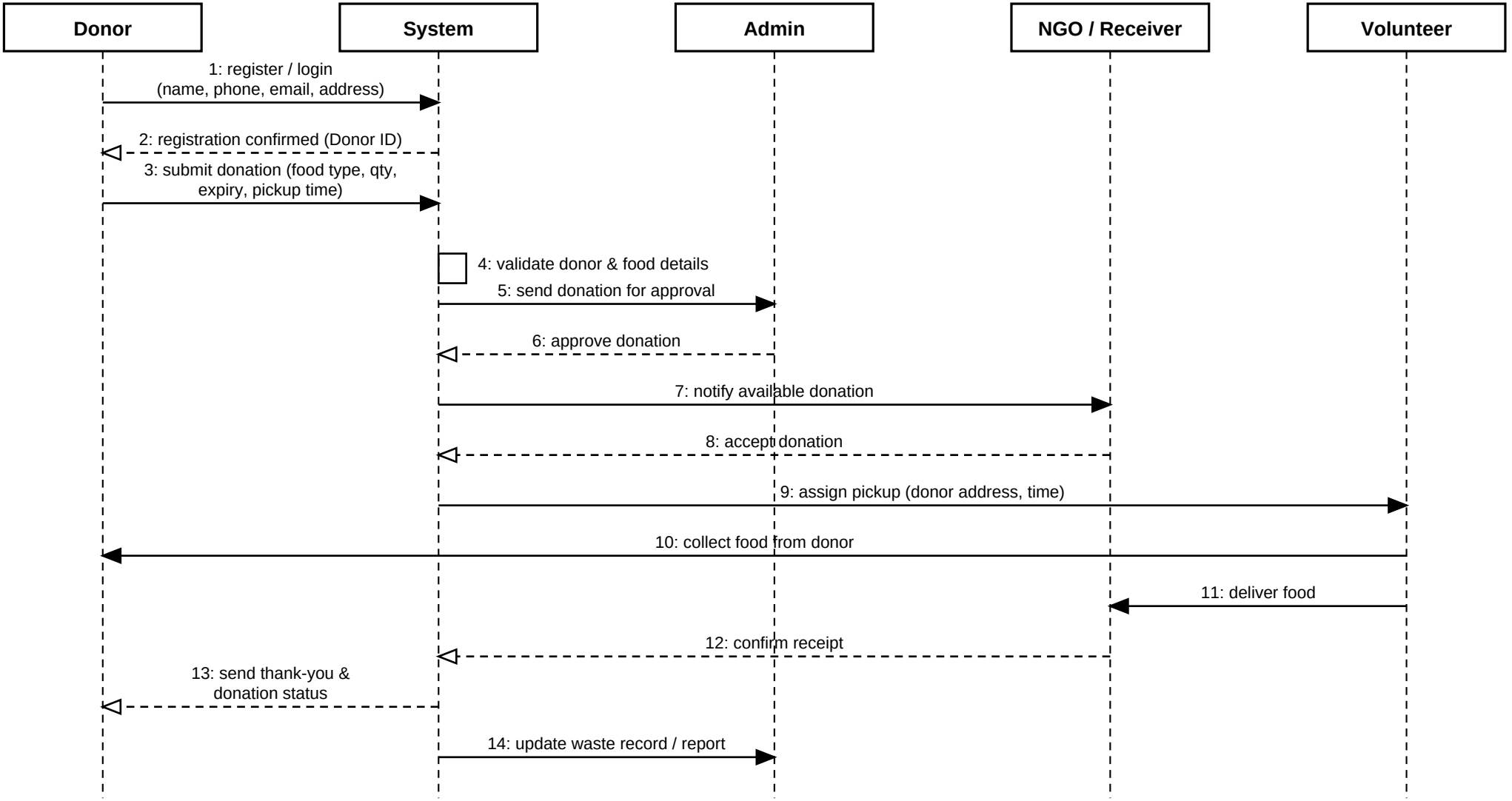
# Use Case Diagram - Food Waste Management System



# Activity Diagram - Food Waste Management System



# Sequence Diagram - Food Waste Management System

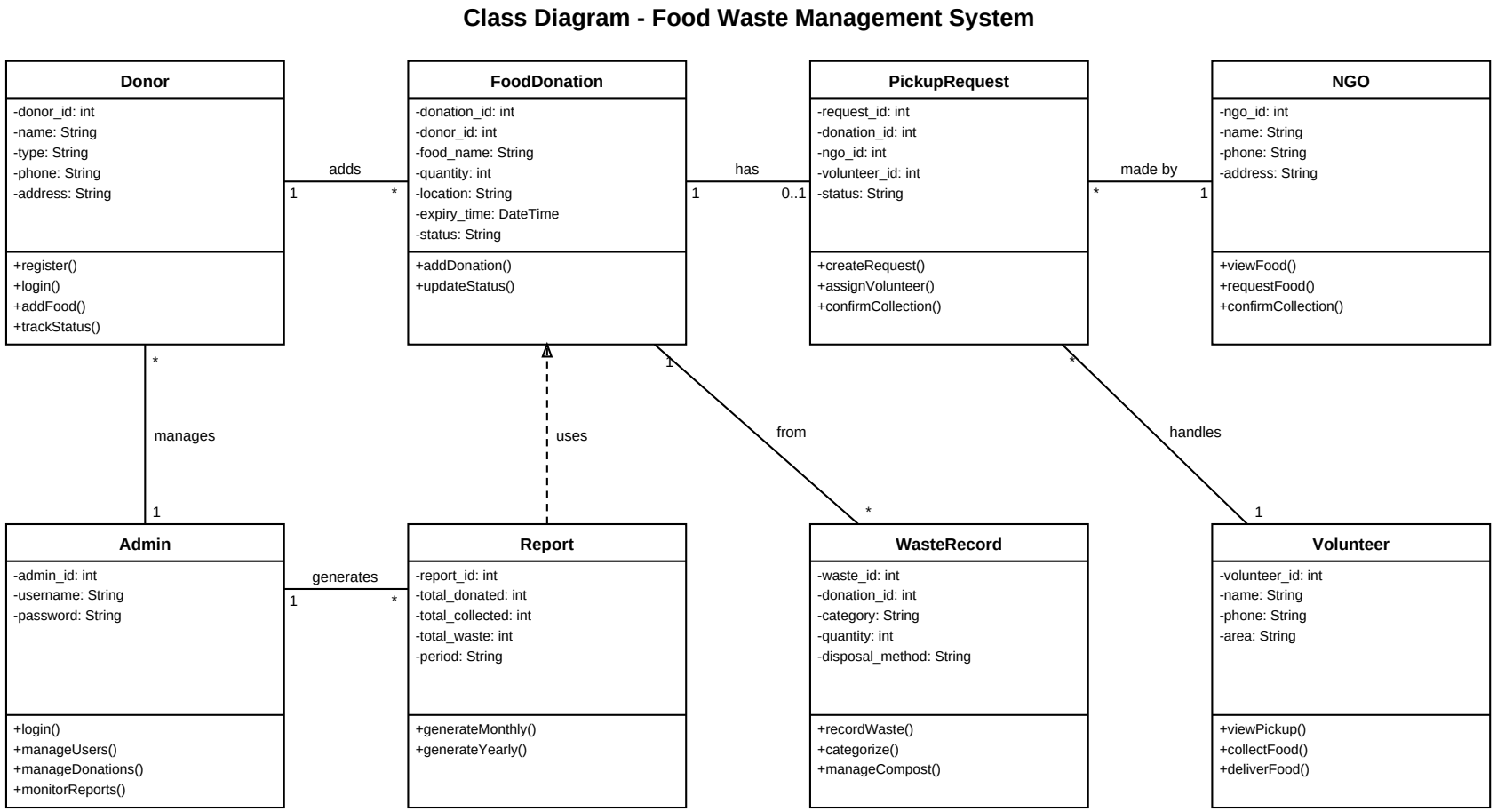



**Donor Details**

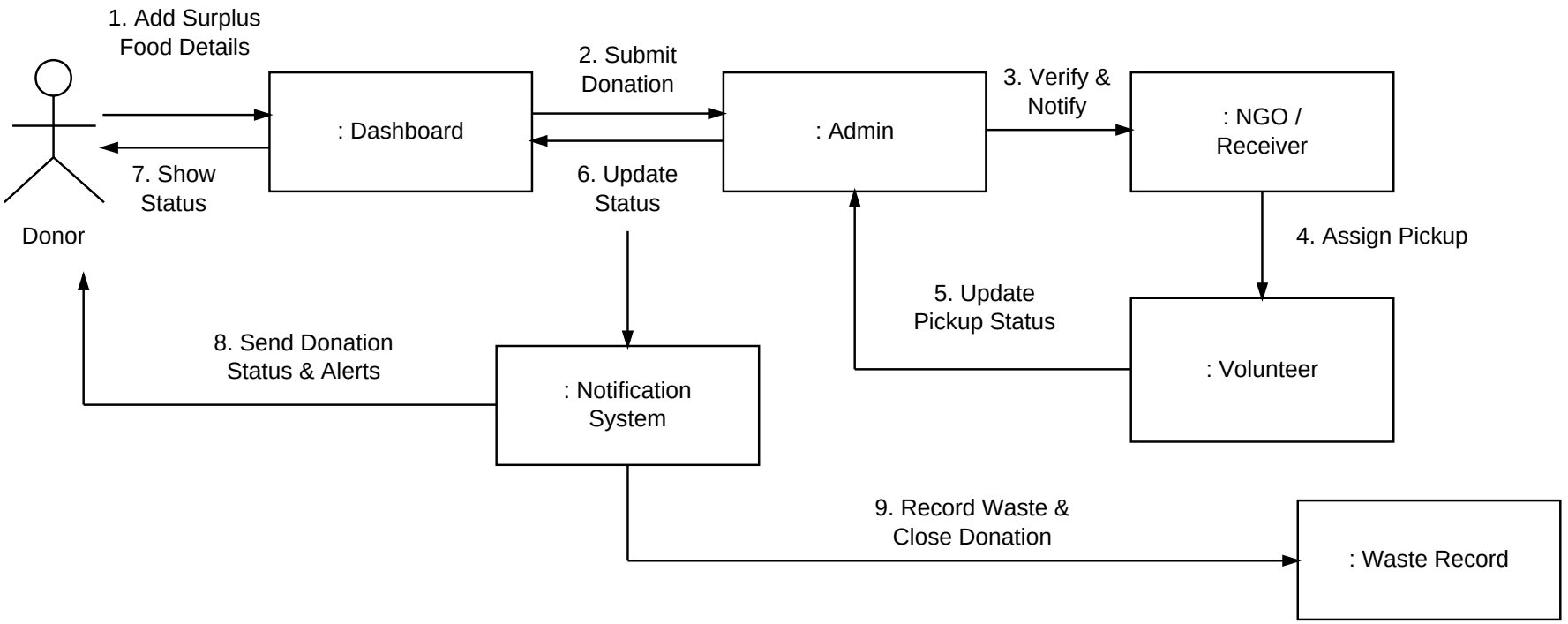
Donor ID | Name (individual / restaurant / event) | Phone | Email | Address (pickup location)

Donation: Food Type | Quantity | Expiry / Prepared Time | Pickup Time | Status (Pending / Approved / Collected / Delivered)

# Class Diagram - Food Waste Management System



# Collaboration Diagram - Food Waste Management System




Collaboration Diagram For Food Waste Management System